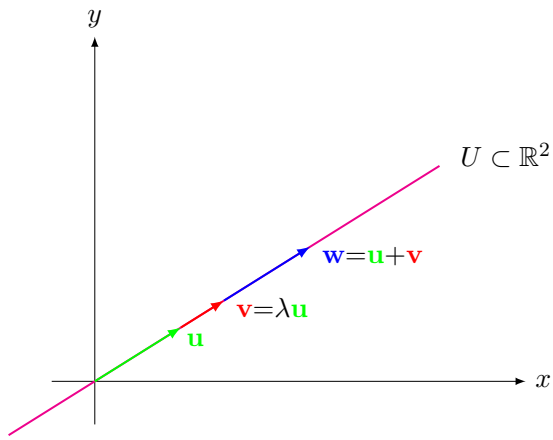

Figure 1: The line U is a linear subspace of $\mathbb{R}^2$

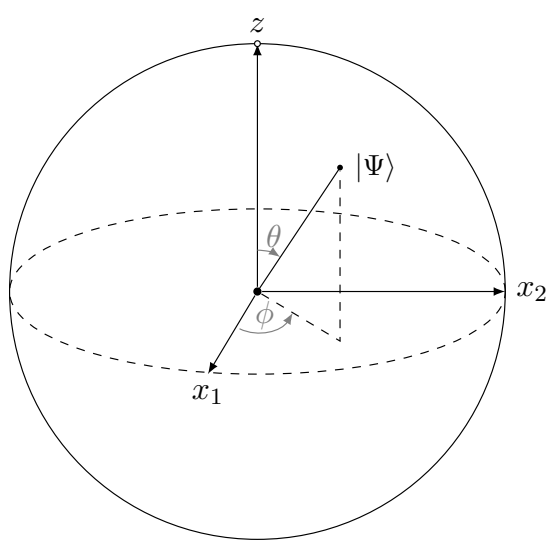

Figure 2: The Bloch Sphere

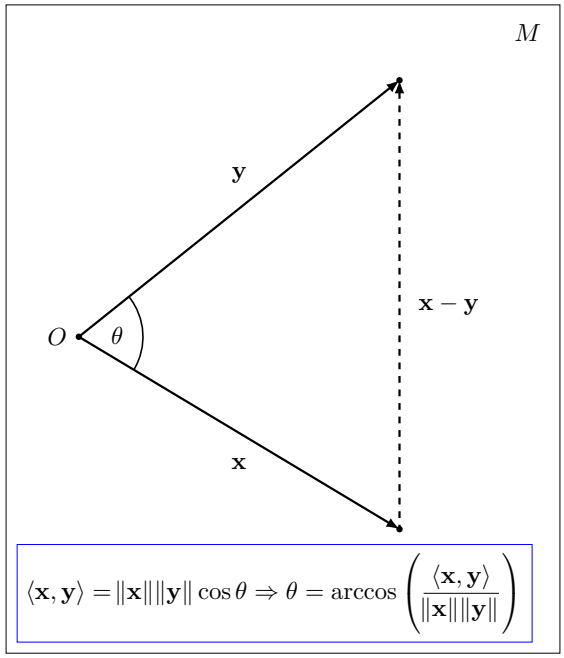

Figure 3: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

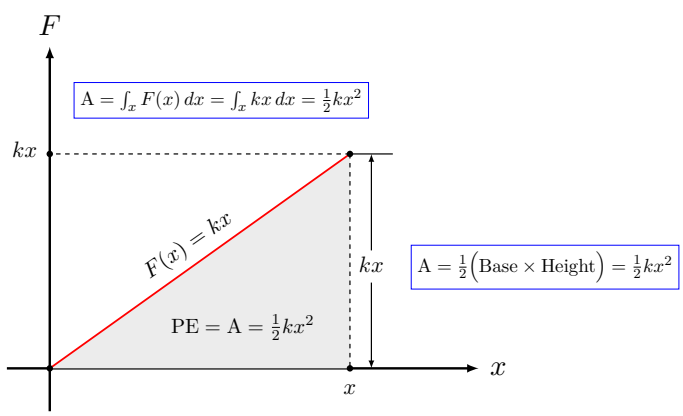

Figure 4: Potential Energy of a Simple Harmonic Oscillator

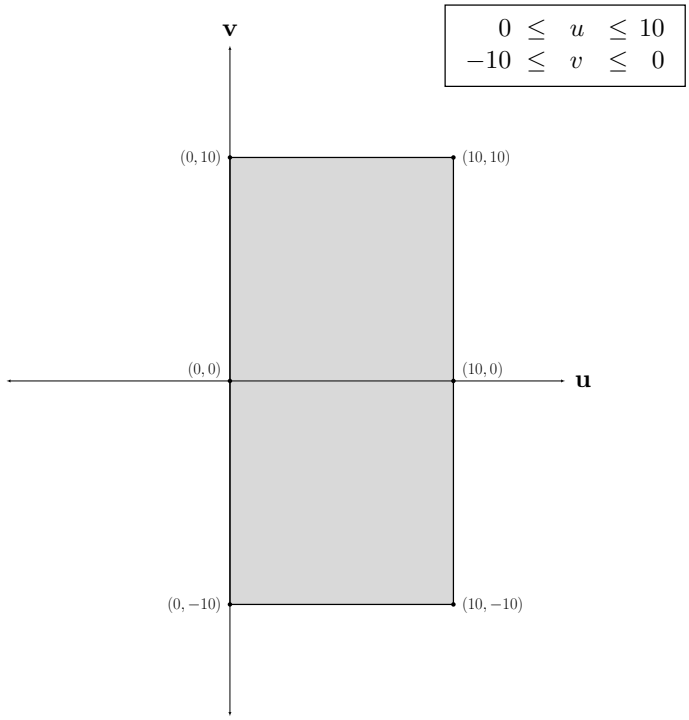

Figure 5: The transformed parallelogram is a rectangle in the uv-plane

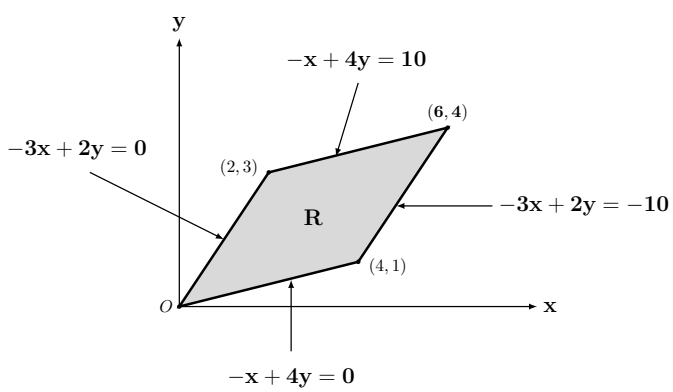

Figure 6: Equations of the Sides of the Parallelogram

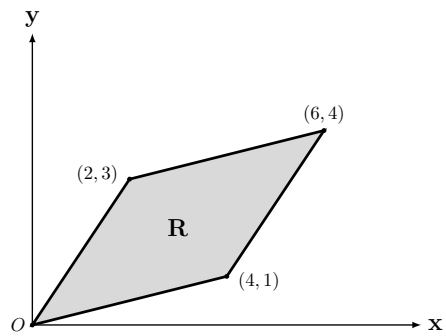

Figure 7: Parallelogram in the xy-plane

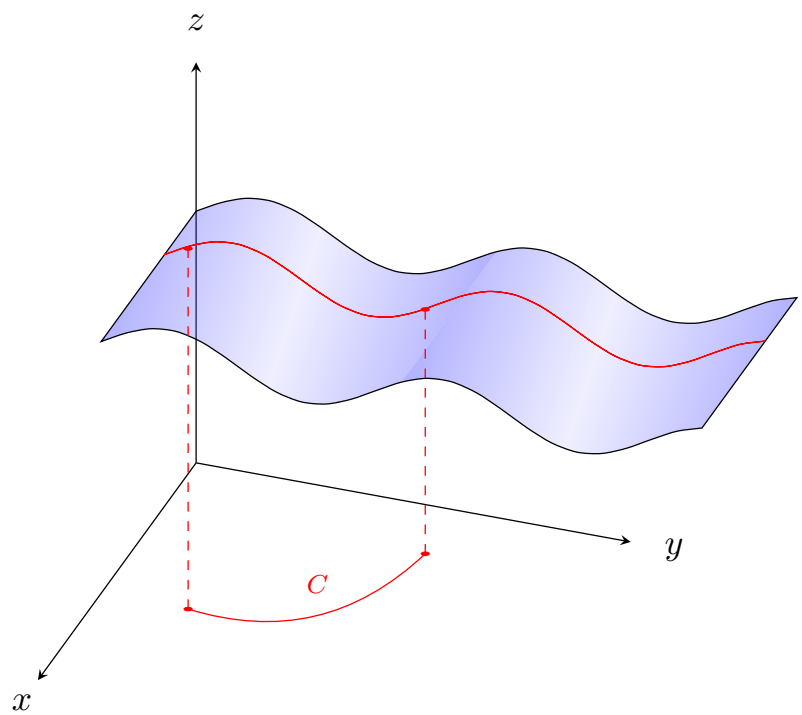

Figure 8: 3D

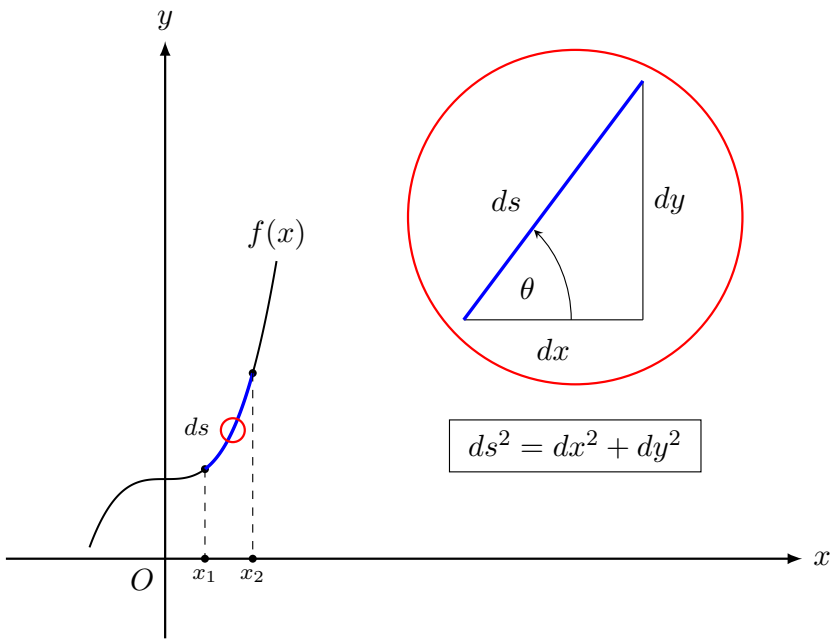

Figure 9: $f(x)$, ds and the Pythagorean Theorem

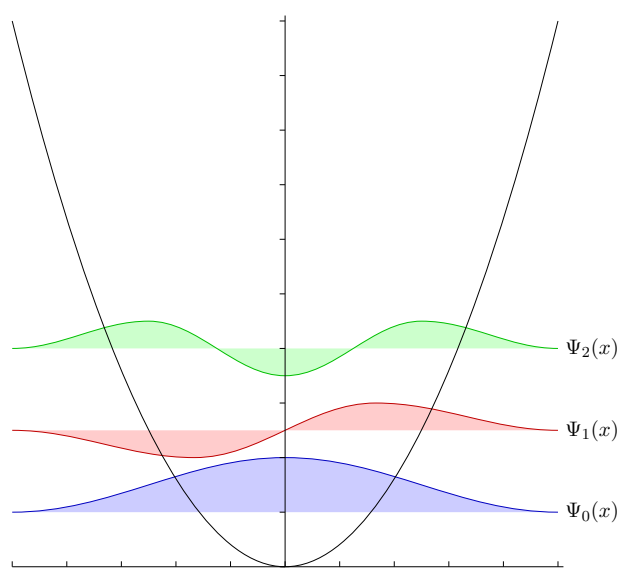

Figure 10: Wave Functions

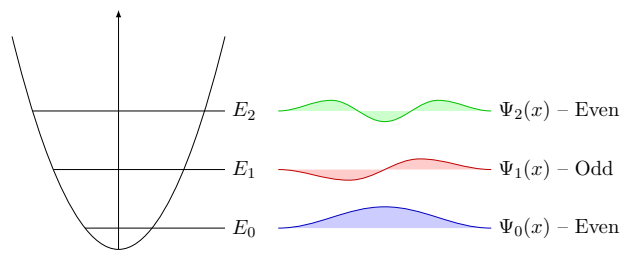

Figure 11: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	81
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 12: A Few Example Kaprekar Numbers

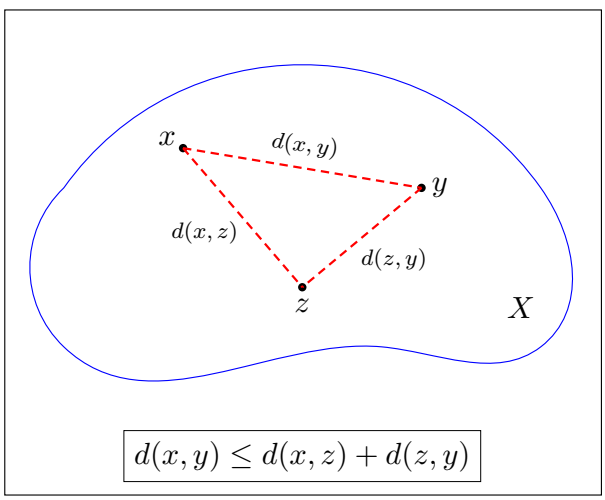

Figure 13: The Triangle Inequality

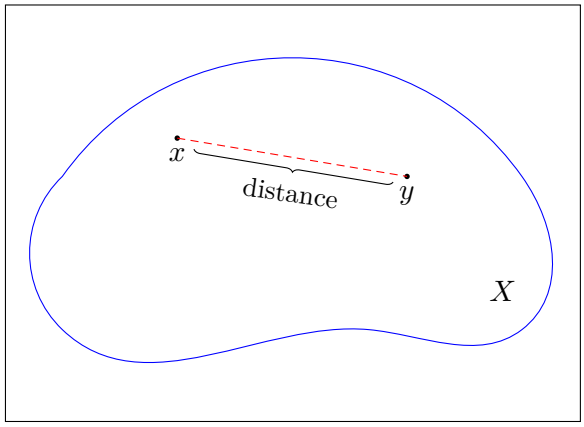

Figure 14: X, x and y

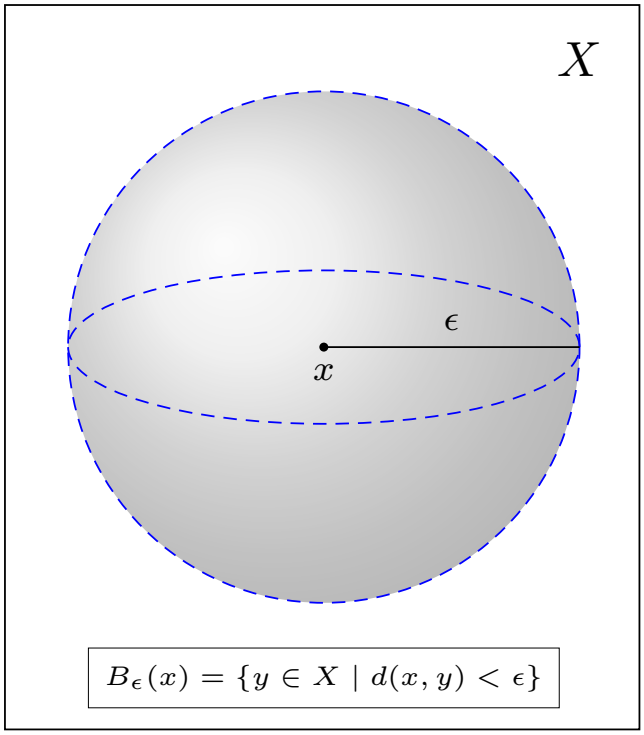

Figure 15: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

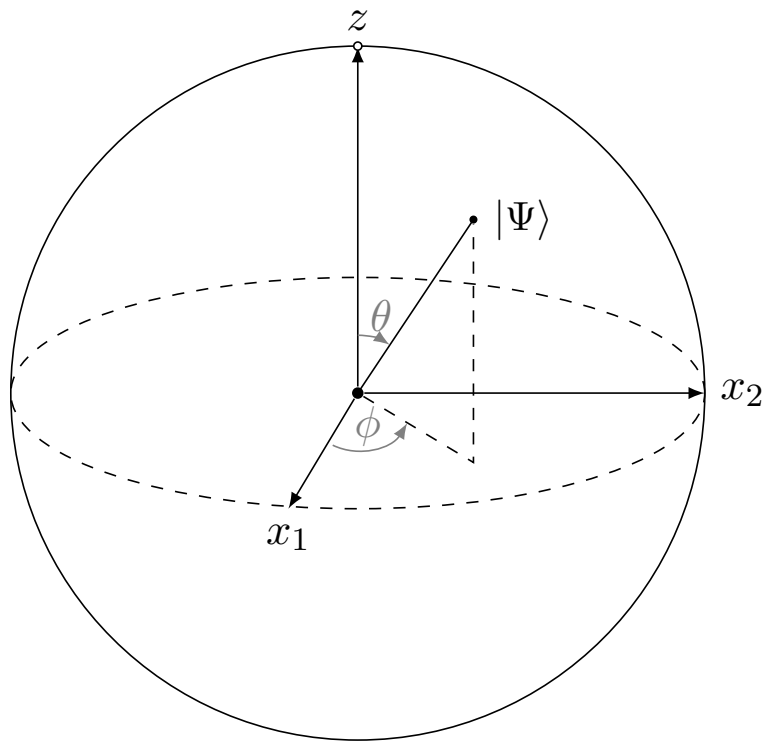

Figure 16: The Bloch Sphere

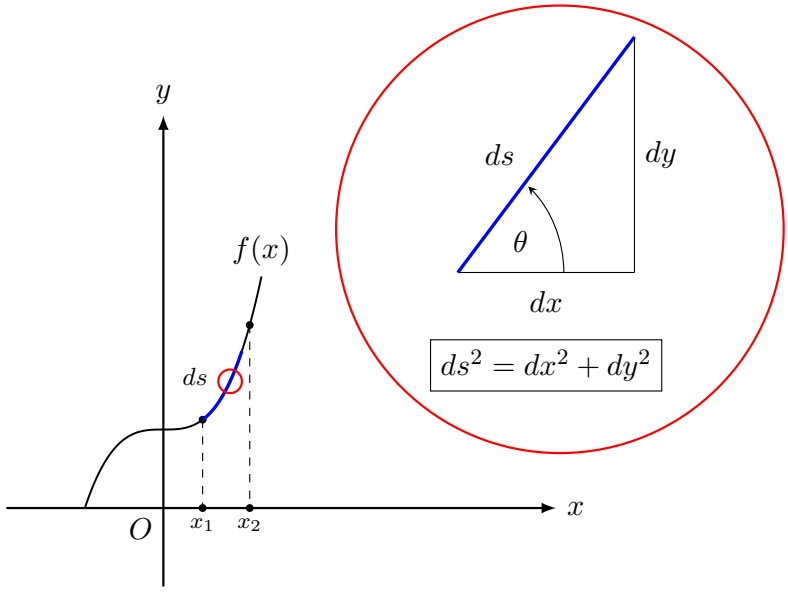

Figure 17: $f(x)$, ds and the Pythagorean Theorem

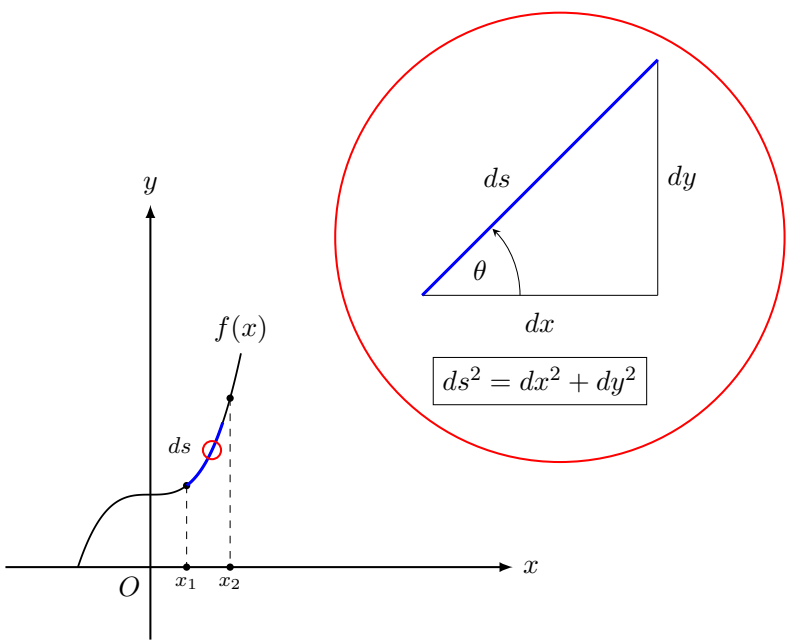

Figure 18: $f(x)$, ds and the Pythagorean Theorem